

Department of Mechanical Engineering

UNIT 2

SOLAR ENERGY

Programme	B.E (UG)
Semester	I/II
Course Name	Renewable Energy Sources
Course Code	ETC144
Course Credits	3-0-0
Unit 2	Solar Energy: Fundamentals; Solar Radiation; Estimation of solar radiation on horizontal and inclined surfaces; Solar radiation Measurements- Pyrheliometers, Pyrometer, Sunshine Recorder. Solar Thermal systems: Flat plate collector; Solar distillation; Solar pond electric power plant. Solar electric power generation- Principle of Solar cell, Photovoltaic system for electric power generation, advantages, Disadvantages and applications of solar photovoltaic system.



UNIT 2

SOLAR ENERGY

Solar energy is defined as the transformation of energy that is present in the sun and is one of the renewable energies. Once the sunlight passes through the earth's atmosphere, most of it is in the form of visible light and infrared radiation. Plants use it to convert into sugar and starches; this conversion process is known as photosynthesis. Solar cell panels are used to convert this energy into electricity.

Solar Energy Advantages and Disadvantages

Advantages of solar energy are:

- **Clean:** It is considered to be the cleanest form of energy as there is no carbon dioxide emission like in the case of fossil fuels which is one of the causes of global warming.
- **Renewable:** There is ample energy available on earth as long as the sun exists.
- **Reliable:** The energy can be stored in the batteries, so there is no unreliability.
- reduction in utility costs.
- Free energy because it can be trapped easily.

Disadvantages of solar energy:

- The production is low during winters and on cloudy days.
- Installation and the initial cost of the materials are expensive.
- Space consumption is more.

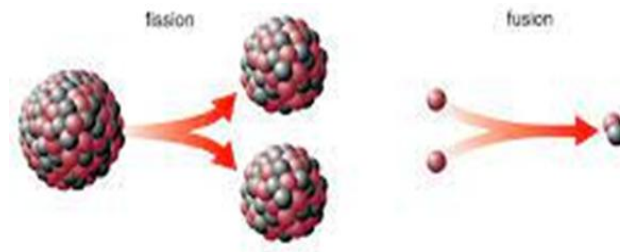
Types of Solar Energy

Solar energy can be classified into two categories depending upon the mode of conversion and type of energy it is converted into. Passive solar energy and active solar energy belong to the mode of conversion and solar thermal energy, photovoltaic solar power and concentrating solar power.

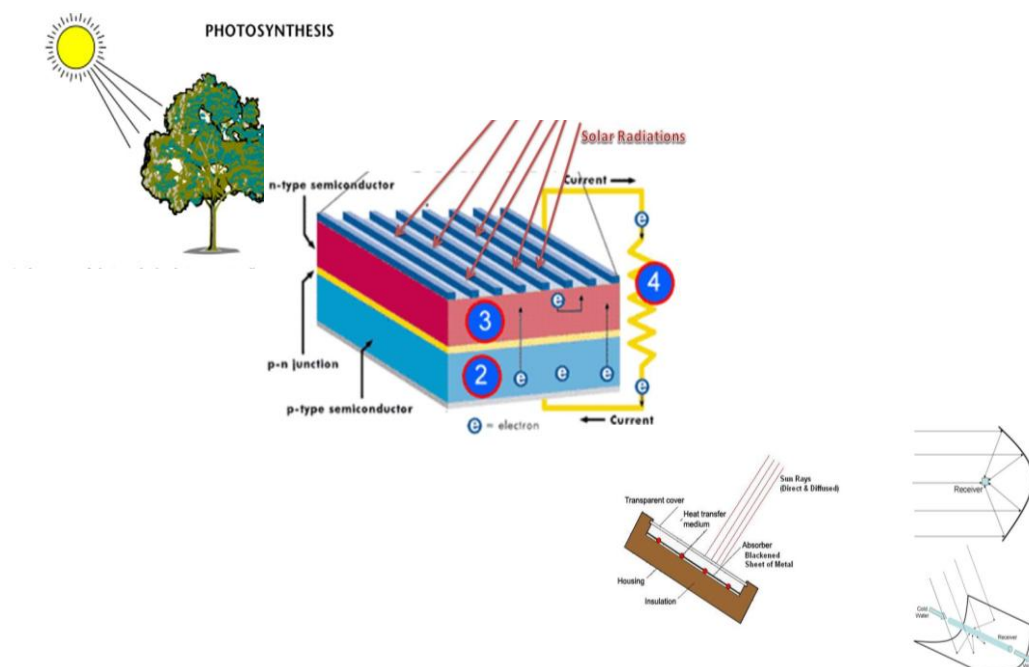
- **Passive solar energy** refers to trapping the sun's energy without using mechanical devices.
- **Active solar energy** uses mechanical devices to collect, store, and distribute energy.
- **Solar thermal energy:** This energy is obtained by converting solar energy into heat.



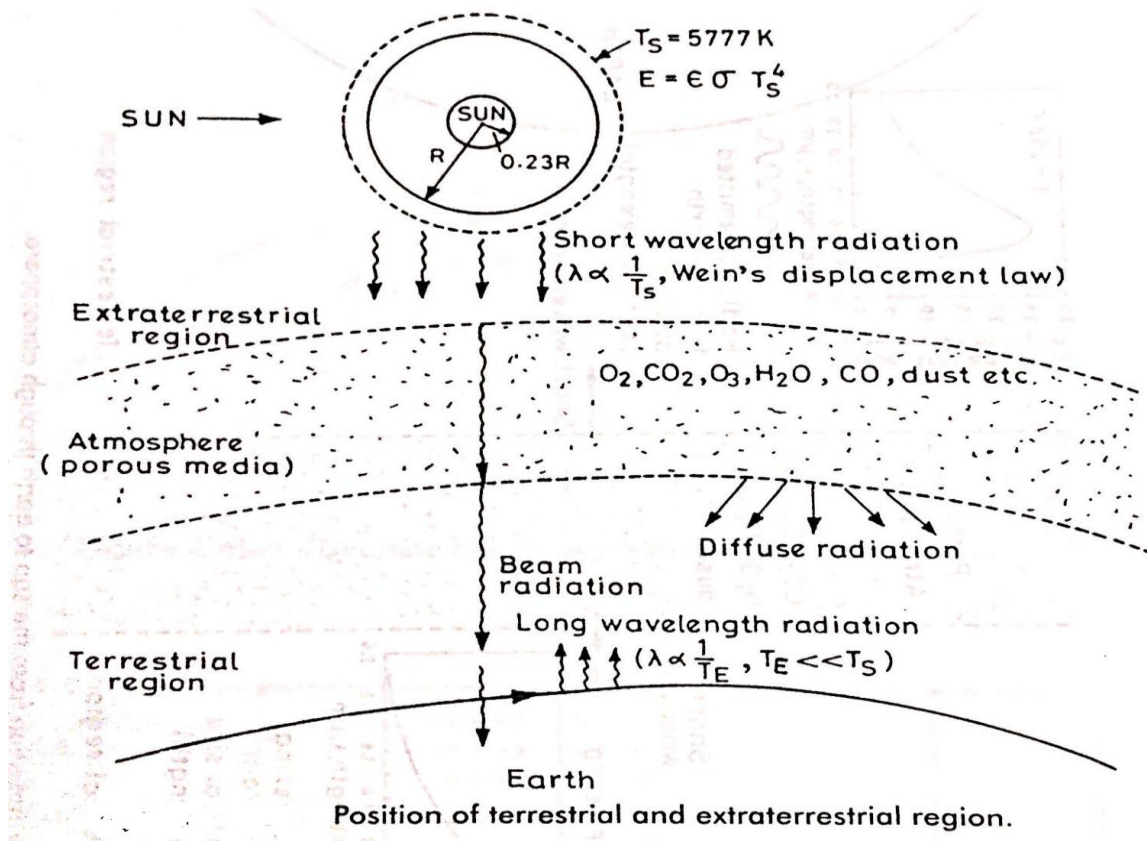
- **Photovoltaic solar power** is the energy obtained by converting solar energy into electricity.
 - **Concentrating solar power:** This is a type of thermal energy used to generate solar power electricity.
- Solar energy Originates with the thermonuclear fusion reactions occurring in the sun.



- Represents the entire electromagnetic radiation (visible light, infrared, ultraviolet, x-rays, and radio waves).
- This energy consists of radiant light and heat energy from the sun.
- Solar energy converted into Helio-chemical, Helio-electrical and Helio-thermal process



POSITION OF TERRESTRIAL AND EXTRATERRESTRIAL REGION



Solar radiations while passing through the earth's atmosphere are subjected to the mechanism of atmospheric absorption and scattering. A fraction of the radiation reaching the earth's surface is reflected back into the atmosphere and is subjected to these atmospheric phenomenon again, the remainder is absorbed by the earth's surface. Absorption occurs due to the presence of water vapour and ozone in the atmosphere and other particulate matter. The sattered radiation redistributed in all directions, some going back into space and some reaching the earth surface.

Fig. shows the position of terrestrial and extraterrestrial regions. The atmosphere absorption is due to Ozone (O_3), Oxygen (O_2), Nitrogen (N_2), Carbon Disioxide (CO_2), Carbon Monoxide(CO) and Water Vapor(H_2O) and, scattering is due to Air molecules, Dust and Water Droplets. The X-rays and extreme Ultra-Violet radiations of the Sun are absorbed highly in the ionosphere by nitrogen, Oxygen and other atmospheric gases: ozone and water vapours largely absorb ultraviolet ($\lambda < 0.4\mu\text{m}$) and infrared radiations ($\lambda > 2.3\mu\text{m}$) respectively. There is almost complete absorption of short wave radiations



($\lambda < 0.29 \mu\text{m}$) in the atmosphere. Hence, the energy in wavelength radiation below $0.29 \mu\text{m}$ and above $2.3 \mu\text{m}$ of the spectra of the solar radiation, incident on the earth's surface is negligible.

➤ Types of radiation

Beam radiation: Solar radiation propagating in a straight line and received at the earth surface without change of direction, i.e., in line with sun is called beam or direct radiation.

Diffuse radiation: Solar radiation scattered by aerosols, dust and molecules is known as diffuse radiation. It does not have a unique direction.

Global radiation: The sum of beam and diffuse radiation is referred to as total or global radiation.

Insolation

Insolation is the amount of solar radiation reaching the earth & also called Incident Solar Radiation.

Solar Constant : The amount of energy received from sun in unit time on unit area perpendicular to the sun's direction and at mean distance of the earth from the sun.

$$I_{on} = I_{sc} \left(1 + 0.033 \cos \frac{360 n}{365} \right)$$

n = day of the year

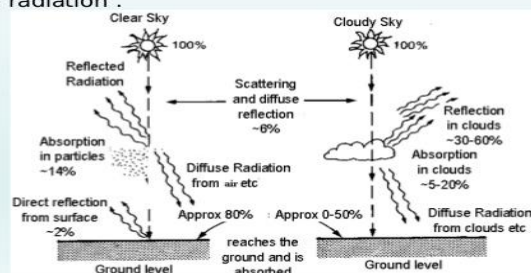
$I_{sc} = 1.353 \text{ kW}$

➤ Atmosphere Influence on Solar Radiation



Atmosphere Influence on Solar Radiation

30% is reflected.
17% is absorbed by the atmosphere.
53% reaches the earth surface:
 31% direct radiation
 22% diffuse radiation

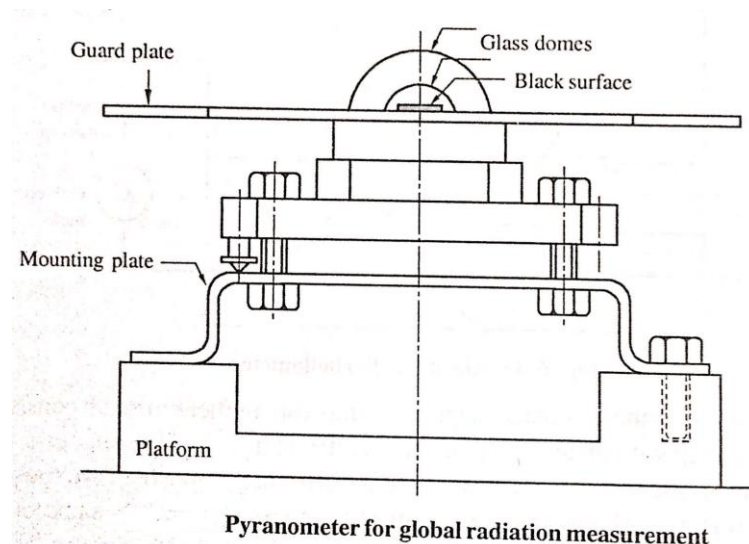


➤ MEASUREMENT OF SOLAR RADIATION

Solar radiation data are measured mainly by the following instruments:

- **Pyranometer:** A pyranometer is designed to measure global radiation, usually on a horizontal surface but can also be used on an inclined surface. When shaded from beam radiation by using a shading ring, it measures diffuse radiation only.
- **Pyrheliometer:** An instrument that measures beam radiation by using a long and narrow tube to collect only beam radiation from the sun at normal incidence.
- **Sunshine recorder** measures the sunshine hours in a day.

Pyranometer:



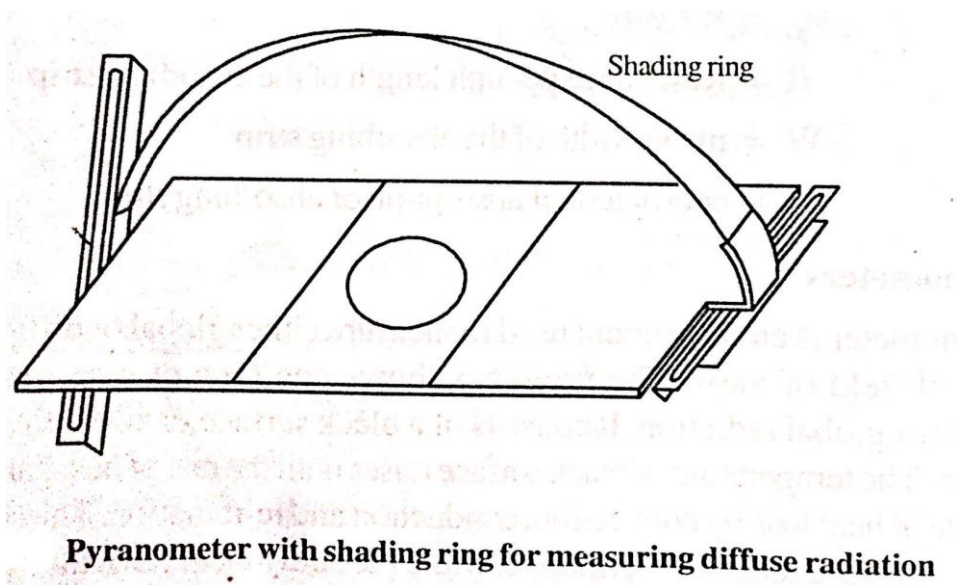
A precision pyranometer is designed to respond to radiation of all wavelengths and hence measures accurately the total power in the incident spectrum. It contains a thermopile whose sensitive surface consists of circular, blackened, hot junctions, exposed to the sun and cold junctions are completely shaded.

The temperature difference between the hot and cold junctions is the function of radiation falling on the sensitive surface. The sensing element is covered by two concentric hemispherical glass domes to shield it from wind and rain. This also reduces the convection currents.

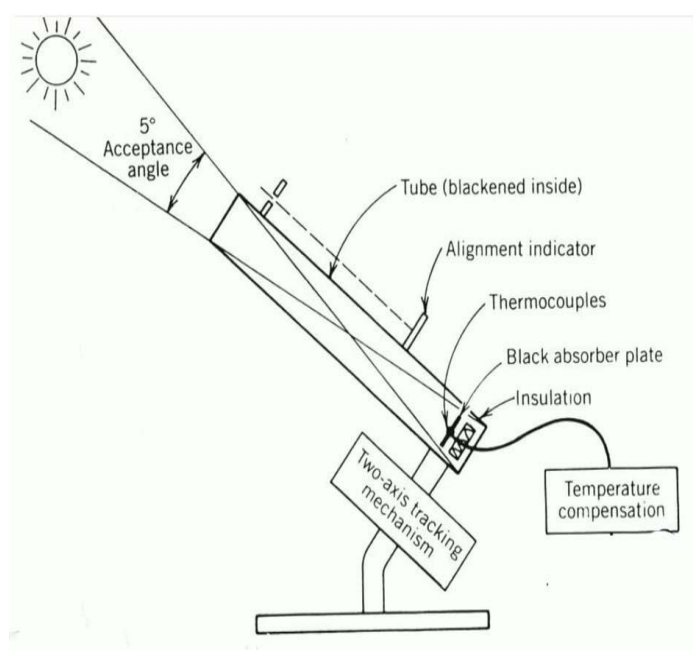
A radiation shield surrounding the outer dome and coplanar with the sensing element, prevents direct solar radiation from heating the base of the instrument. The instrument

has a voltage output of approximately $9 \mu\text{V}/\text{W}/\text{m}^2$ and has an output impedance of 650 Ω .

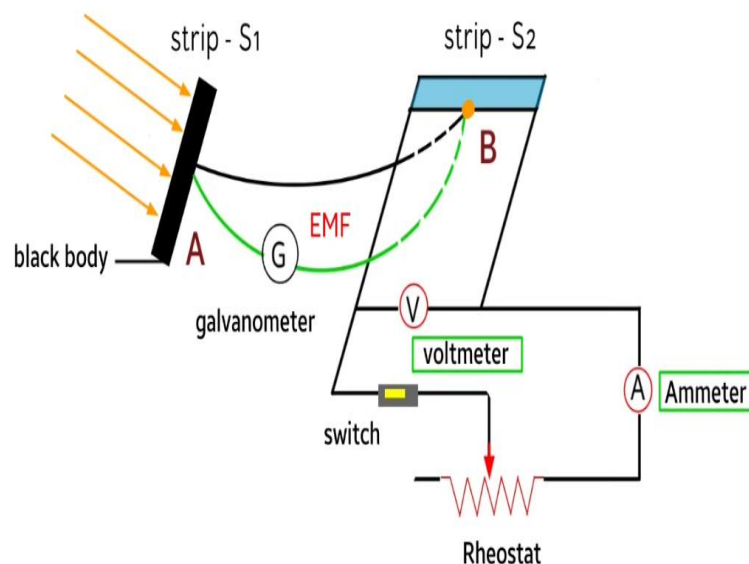
The pyranometer, when provided with a shadow band (or occulting disc) to prevent beam radiation from reaching the sensing element, measures the diffuse radiation only. Such an arrangement of shadow bandstand (model: SBS) is shown below



Pyrheliometer: An instrument which measures beam radiation



- The normal incidence pyrheliometer uses a long collimator tube to collect beam radiation whose field of view is limited to a solid angle of 5.5° (generally) by appropriate diaphragms inside the tube.
- The inside of the tube is blackened to absorb any radiation incident at angles outside the collection solid angle. At the base of the tube a wire wound thermopile having a sensitivity of approximately 8 mV/W/m^2 and an output impedance of approximately 200 W is provided.
- The tube is sealed with dry air to eliminate absorption of beam radiation within the tube by water vapour. A tracker is needed if continuous readings are desired.



- The circuit diagram of the pyrheliometer is shown above. It includes two equal strips specified with two strips S_1 & S_2 with area 'A'. Here, a thermocouple is used where its one junction can be connected to S_1 whereas the other is connected to S_2 . A responsive galvanometer can be connected to the thermocouple.
- The S_2 Strip is connected to an exterior electrical circuit. Once both the strips are protected from the radiation of solar, then the galvanometer illustrates there is no deflection because both the junctions are at equal temperature.
- When S_1 strip gets heat radiations from the sun, then strip temperature will be increased, thus the galvanometer illustrates deflection. When current is supplied



throughout the S₂ strip, then it is adjusted and the galvanometer illustrates there is no deflection.

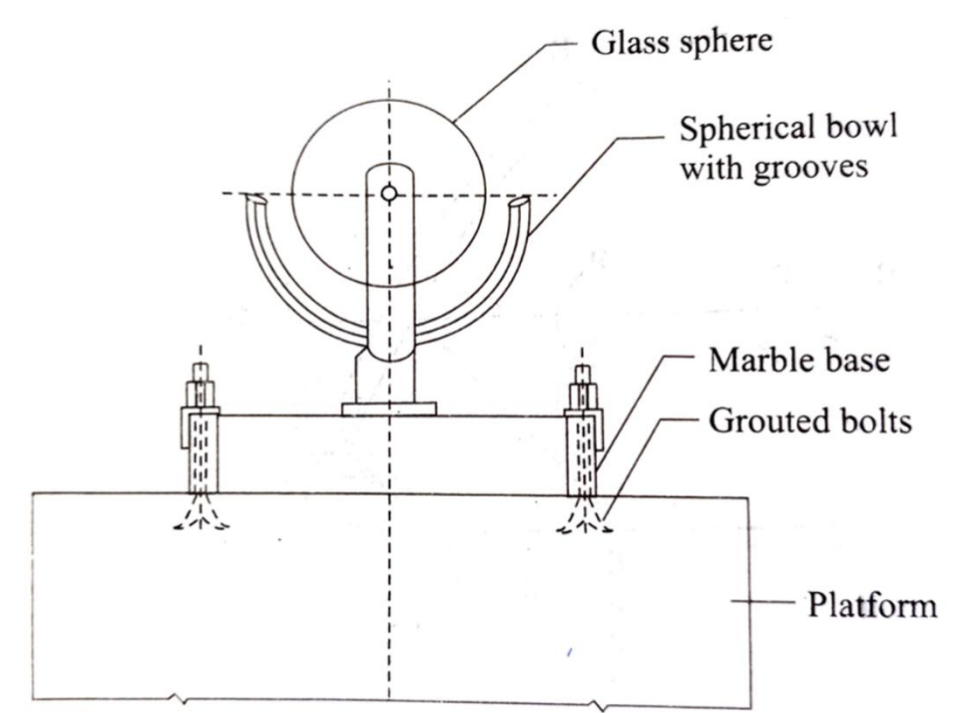
- Now, again both the strips are at equal temperature. If the heat radiation amount occurred over the unit area within the unit time on S₁ strip is 'Q' & its absorption co-efficient, so the heat radiation amount which is absorbed through the strip S₁ within unit time is 'QA'.
- In addition, the heat generated in unit time within the strip S₂ can be given through VI. Here, 'V' is the potential difference & 'I' is the flow of current through it. When heat absorbed is equivalent to the heat generated, so

$$QA = VI$$

$$Q = VI/A$$

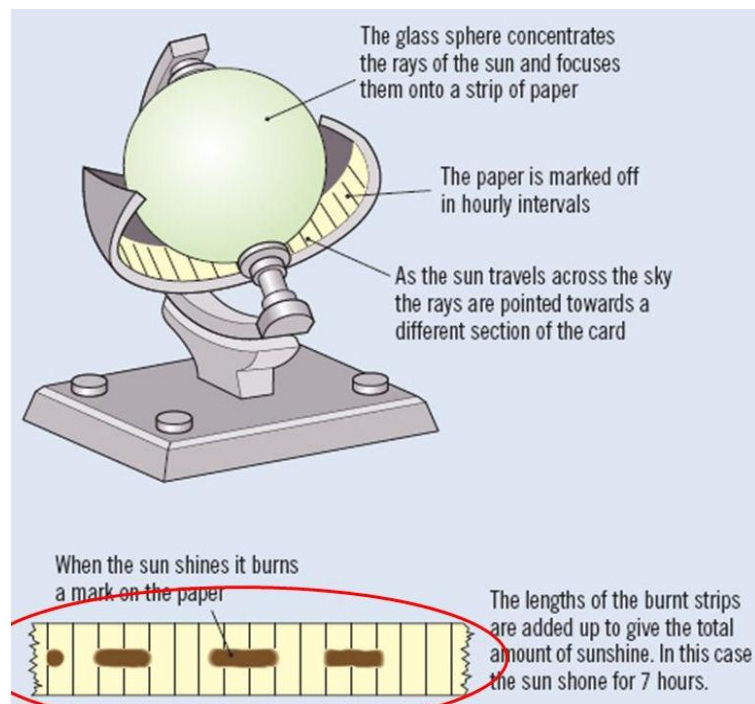
By substituting the values of V, I & A the value of 'Q' (Radiation heat) can be calculated.

Sunshine Recorder:





- This instrument measures the duration in hours, of bright sunshine during the course of the day. It essentially consists of glass sphere (about 10 cm in diameter) mounted on its axis parallel to that of earth, within a spherical section (bowl).
- The bowl and glass sphere is arranged in such a way that sun's rays are focused sharply at a spot on a card held in a groove in the bowl. The card is prepared from special paper bearing a time scale. As the sun moves, the focused bright sunshine burns a path along this paper.
- The length of the trace thus obtained on the paper is the measure of the duration of the bright sunshine. Three overlapping pairs of grooves are provided in the spherical segment to take care of the different seasons of the year.





➤ ESTIMATION OF SOLAR RADIATION ON HORIZONTAL AND INCLINED SURFACES

Symbols Used

$H_{g/d}$ - Daily global solar radiation on horizontal surface

$\bar{H}_{g/d}$ - Monthly averaged daily global solar radiation on horizontal surface

H_o - Daily extra-terrestrial global solar radiation

$\bar{H}_o$ - Monthly averaged extra-terrestrial global solar radiation

$I_{g/d}$ - Daily hourly solar radiation on horizontal surface

$\bar{I}_{g/d}$ - Monthly averaged hourly global solar radiation on horizontal surface

Estimating solar radiation empirically: Global radiation

Estimation of **monthly average global radiation**
on **horizontal surface**

$$\frac{\bar{H}_{ga}}{\bar{H}_{oa}} = a + b \left(\frac{\bar{S}_a}{S_{\max a}} \right)$$

Where,

H_{ga} → monthly averaged daily global radiation on a horizontal surface

H_{oa} → monthly averaged extra-terrestrial solar radiation at horizontal surface (at top of atmosphere)

S_a and $S_{\max a}$ → monthly averaged daily sunshine hours and maximum possible daily sunshine hours (the day length) at a given location.

a and **b** → constant



Table 2.7.1. Constants a and b for Some Indian Towns

<i>Location</i>	<i>a</i>	<i>b</i>	<i>Location</i>	<i>a</i>	<i>b</i>
Ahmedabad	0.28	0.48	Manglore	0.27	0.43
Bangalore	0.18	0.64	Shillong	0.22	0.57
Baroda	0.28	0.48	Srinagar	0.35	0.40
Bhavnagar	0.28	0.47	Trivandrum	0.37	0.39
Bhopal	0.27	0.50	Vishakhapatnam	0.28	0.47
Calcutta	0.28	0.42	Nagpur	0.27	0.50
Goa	0.30	0.48	New Delhi	0.25	0.57
Jodhpur	0.33	0.46	Poona	0.31	0.43
Madras	0.30	0.44	Roorkee	0.25	0.56

Monthly averaged daily Diffuse radiation- horizontal surface

Studies on Indian solar radiation data

$$0.3 \leq K_T \leq 0.7$$

$$\frac{H_{da}}{H_{ga}} = 1.411 - 1.696 K_T$$

One other analysis presented following

$$\frac{H_{da}}{H_{ga}} = 1.354 - 1.570 K_T$$

- K_T = sky monthly averaged clearness index, $= H_{ga} / H_{oa}$
- Typically diffuse radiation is about 10 to 20% of the global radiation on horizontal surface
- Diffuse radiation component in India is larger than Europe



Radiation on tilted south facing surface

Since, the solar radiation falling horizontal surface,

$$I_T = I_b + I_d + I_r$$

The solar radiation falling on tilted surface is given by

$$I_T = I_b r_b + I_d r_d + (I_b + I_d) r_r$$

Where, r_b , r_d , and r_r are the tilt factors for beam, diffuse and reflected radiations respectively

$$\text{Tilt Factor} = \frac{\text{Radiation flux falling on tilted surface}}{\text{Radiation flux falling on horizontal surface}}$$

SOLAR THERMAL SYSTEMS:

1. Flat Plate Collector
2. Solar Distillation
3. Solar Pond Electric Power Plant

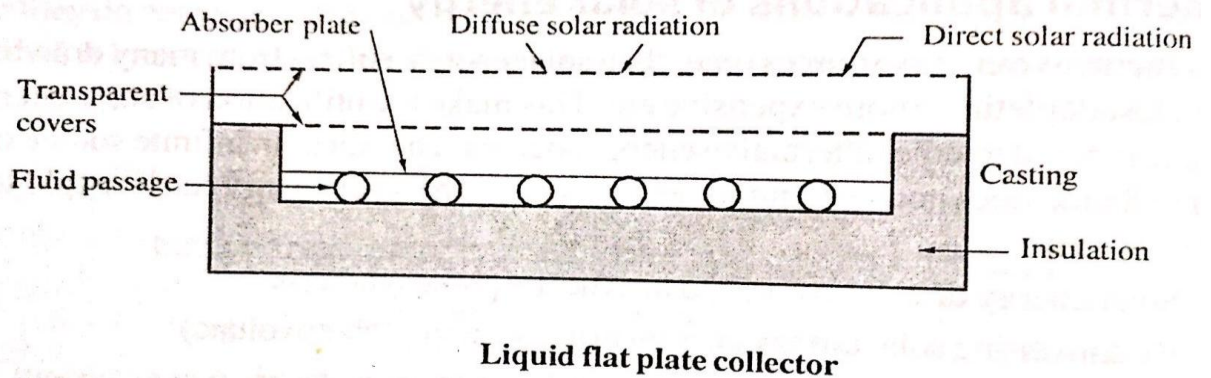
1. FLAT PLATE COLLECTOR

1. Liquid flat plate collector
2. Solar air heater

1. Liquid flat plate collector

A Flat plate collector is a solar panel device that uses solar energy to generate thermal energy. It converts solar power into thermal energy, i.e., cheaper energy utilising water as an operating fluid.

A Flat plate solar collector takes in solar radiation and transmits heat to the functioning medium. It is suitable for several thermal applications. The average temperature range of FPC devices is 100° C. Besides, these devices have an economical cost of investment.



CONSTRUCTION

Here are the typical components of a flat plate collector:

Absorbing Plate: It is a component inside the collector that traps solar radiation. The absorbing plate converts the solar power into thermal power. It is a dark plate, generally made of copper foil and area of 1.7 to 2.9 m² with 1 to 2mm thick.

Tubes or Passages: The absorbing plate in a flat plate collector has a grid of conduits. These are tubes or passages for circulation of the heat transfer fluid. It aids in conducting the working fluid from the inlet to the outlet and diameter of 1 to 1.5cm. The distance between pipe 5 to 15cm.

Insulation & Glazing Cover: Insulation minimises heat loss from the absorbing plate. It is present on the sides & back of the plate collector. The insulation is typically made up of synthetic foam sheets (fibreglass, polyurethane, etc.) and having thickness of 5 to 10cm.

Casing or Container: The casing acts as a resting surface or closure for the components of the flat plate solar collector. It is made up of aluminium and holds all elements in place.

WORKING

The working of a flat plate collector (FPC) involves the transfer of heat or thermal energy. The operating medium exchanges heat from the sun's rays.

The heat-absorbing plate of the collector is exposed to sunlight. As the sun rays hit the flat plate surface, a portion of their energy is transformed into heat. This leads to a rise in the temperature of the flat plate solar collector.

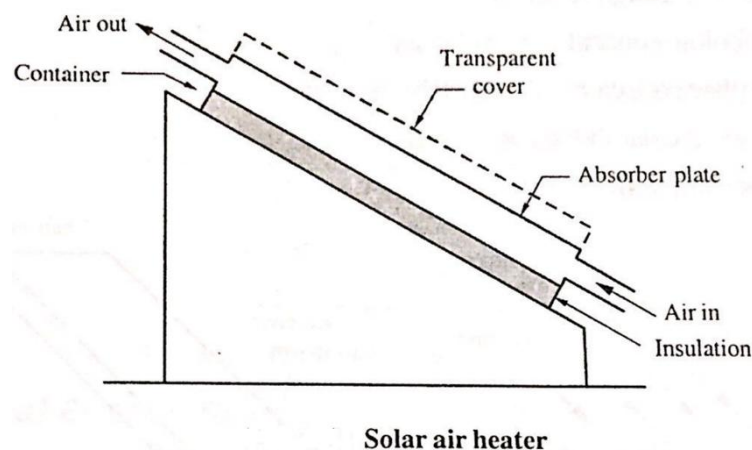


When a fluid is passed inside the collector, the temperature of the fluid increases as the heat from the absorbing plate heat is transmitted to the fluid.

Eventually, the fluid transmits the thermal energy from collectors to the functioning energy systems for different uses. It works on the principles of the 1st & 2nd Laws of Thermodynamics.

2. Solar air heater

Solar air heating technologies use only free, renewable, and clean energy, and can help defray the rising cost of conventional energy. Solar air heating systems absorb thermal energy from direct sunlight to heat air; this heated air can then be circulated through buildings to provide heat.



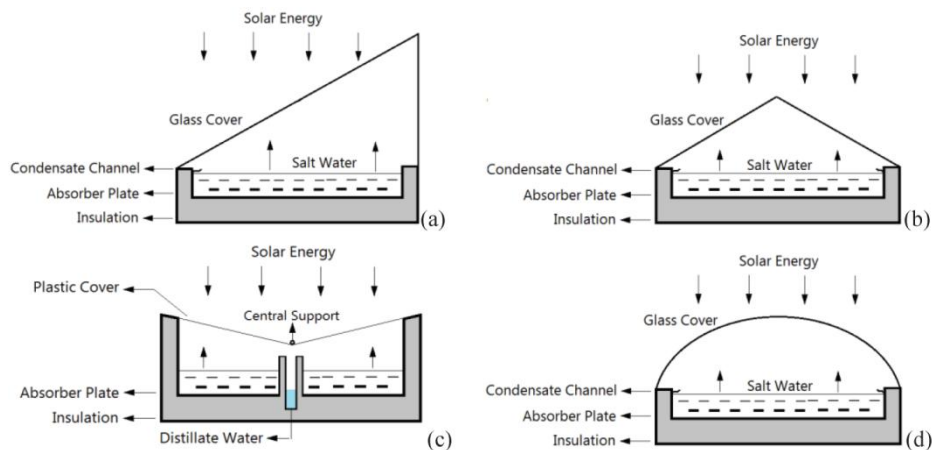
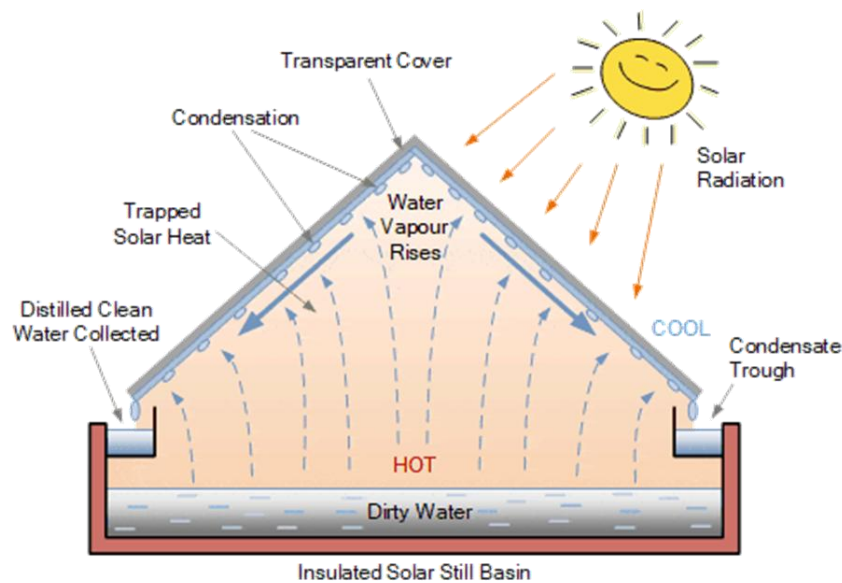
WORKING

A simple solar air collector consists of a flat, dark metal absorber plate encased in an airtight, insulated metal frame with glass over the top. The more air moves, the warmer it gets, so more complex designs use roughened surfaces or channels to increase the disturbance of the air as it circulates through the collector. Commercial manufacturers use black aluminum for the absorber plate, but one DIY design employs aluminum drink cans that have been painted black. It's a low-cost, environmentally friendly solution that produced modest results for the innovative designer

2.SOLAR DISTILLATION

There is a great need to find ways to supply water for the Earth's population. Many countries are facing water shortages and/or have residents who use and drink

contaminated water. Finding different ways to use our renewable resources (for example, solar power) has become an interest. Solar water distillation is the process of using energy from the sunlight to separate freshwater from salts or other contaminants. The untreated water absorbs heat, slowly reaching high temperatures. The heat causes the water to evaporate, cool, and condense into vapour, leaving the contaminants behind. Solar stills can be used for low capacity and self-reliant water supplying systems.



WORKING

Solar water distillers or solar stills are usually used in remote areas where there is limited access to freshwater. The basic principles of solar water distillation are simple, yet effective, as distillation replicates the way nature makes rain. A solar still works on two scientific



principles: evaporation and condensation. The salts and minerals do not evaporate with the water. For example, table salt does not turn into vapour until it gets to a temperature over 1400°C . However, it still does take a certain amount of energy for water to turn into water vapour. While a certain amount of energy is needed to raise the temperature of a kilogram of water from 0°C to 100°C , it takes five and one-half times that much to change it from water at 100°C to water vapour at 100°C . Practically all this energy, however, is given back when the water vapour condenses.

Most stills are simple black bottomed vessels filled with water and topped with clear glass or plastic. Sunlight that is absorbed by the black material speeds the rate of evaporation. The evaporation is then trapped by the clear topping and funneled away. Most pollutants do not evaporate, so they are left behind. Most stills need to be about six square meters in size to produce enough water for a single person for a day. Multiple solar distillation systems are required to produce a large quantity of distilled water.

Advantages:

- It is a relatively cheap and low-maintenance system.
- It can be used at the household level and scaled up through programmatic approaches.
- There are climate change adaptation and mitigation benefits.
- There are no energy costs.
- There are no moving parts.

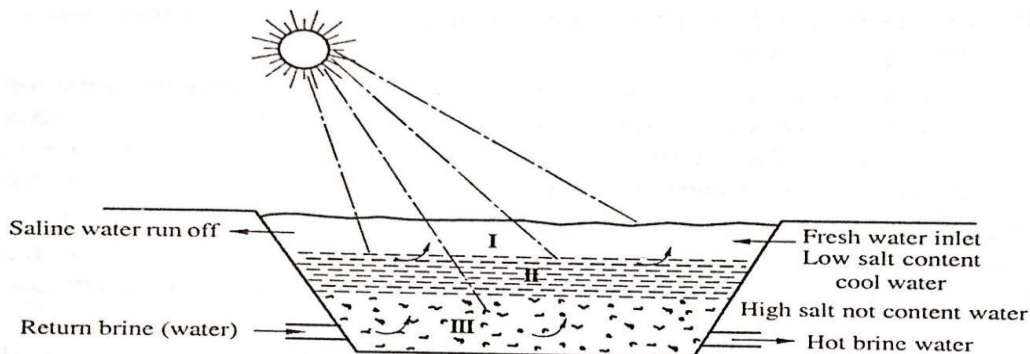
Disadvantages:

- Rate of distillation is usually very slow (6 litres of water per sunny day).
- It is not suitable for larger consumptive needs.
- The materials required for the distiller may be difficult to obtain in some areas.

3.SOLAR POND ELECTRIC POWER PLANT

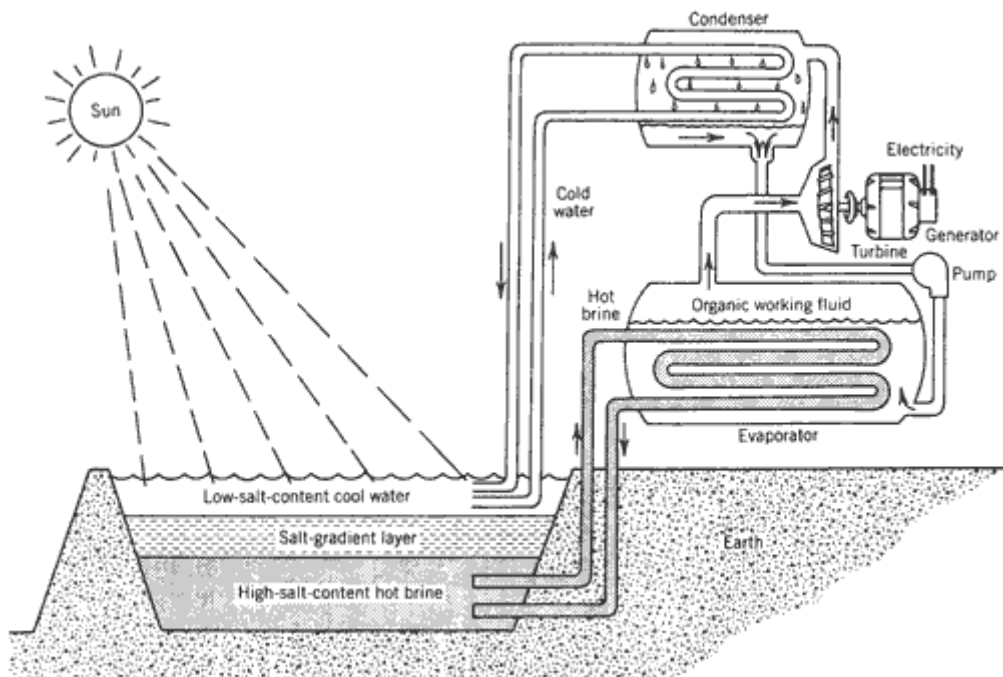
A **solar pond** is a solar energy collector, generally fairly large in size, that looks like a pond. This type of solar energy collector uses a large, salty lake as a kind of a flat plate collector that absorbs and stores energy from the Sun in the warm, lower layers of the pond. These

ponds can be natural or man-made, but generally speaking the solar ponds that are in operation today are artificial.



- I - Surface or upper convective zone (UCZ)
 - II - Non convective or Intermediate gradient zone.
 - III - Storage zone or lower convective zone.
- Salt dissolved in water - magnesium chloride, sodium chloride or sodium nitrate.

Principle of solar pond



WORKING

The key characteristic of solar ponds that allow them to function effectively as a solar energy collector is a salt-concentration gradient of the water. This gradient results in water that is heavily salinated collecting at the bottom of the pond, with concentration decreasing towards the surface resulting in cool, fresh water on top of the pond. This collection of salty water at



the bottom of the lake is known as the "storage zone", while the freshwater top layer is known as the "surface zone". The overall pond is several meters deep, with the "storage zone" being one or two meters thick.

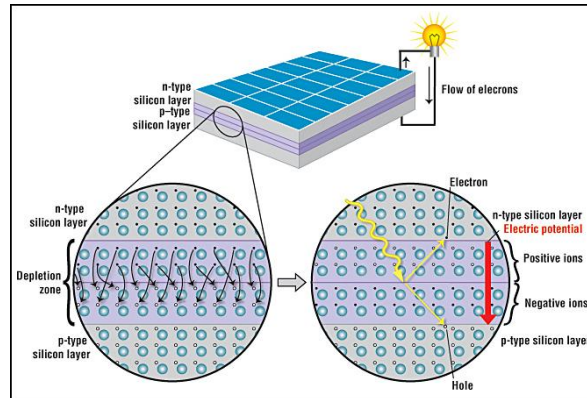
These ponds *must* be clear for them to operate properly, as sunlight cannot penetrate to the bottom of the pond if the water is murky. When sunlight is incident on these ponds, most of the incoming sunlight reaches the bottom and thus the "storage zone" heats up. However, this newly heated water cannot rise and thus heat loss upwards is prevented. The salty water cannot rise because it is heavier than the fresh water that is on top of the pond, and thus the upper layer prevents convection currents from forming. Because of this, the top layer of the pond acts as a type of insulating blanket, and the main heat loss process from the storage zone is stopped. Without a loss of heat, the bottom of the pond is warmed to extremely high temperatures - it can reach about 90°C. If the pond is being used to generate electricity this temperature is high enough to initiate and run an organic Rankine cycle engine.

SOLAR ELECTRIC POWER GENERATION

- Low Temperature System
- Medium Temperature System
- High Temperature System
- **Photovoltaic Conversion**

PRINCIPLE OF SOLAR CELL

When sunlight strikes a solar cell, electrons in the silicon are ejected, which results in the formation of "holes"—the vacancies left behind by the escaping electrons. If this happens in the electric field, the field will move electrons to the n-type layer and holes to the p-type layer. If you connect the n-type and p-type layers with a metallic wire, the electrons will travel from the n-type layer to the p-type layer by crossing the depletion zone and then go through the external wire back of the n-type layer, creating a flow of electricity.



Different Types of PV Cells

1. Crystalline Silicon PV Cells (Monocrystalline)

- Solar Cells are manufactured from [crystalline silicon](#).
- To make solar cells out of silicon, manufactured silicon crystals are sliced to about 300 micrometers thick and coated to work as a semiconductor to capture solar energy.

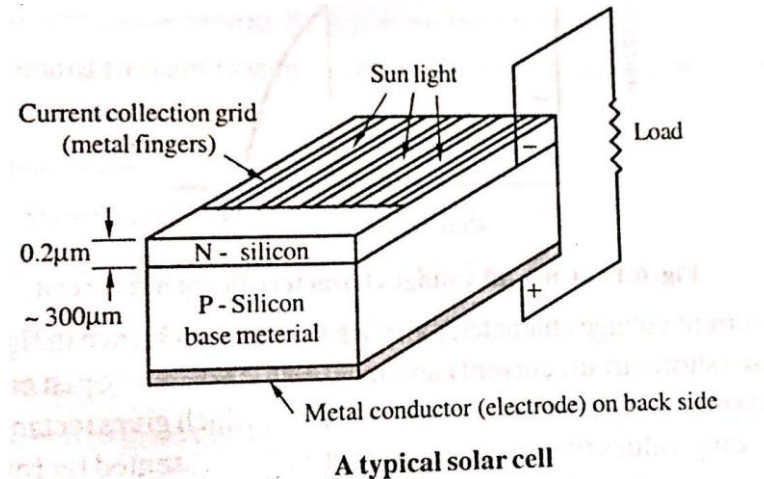
2. Thin-film or Polycrystalline PV Cells

- Thin-film PV cells use amorphous silicon or an alternative to silicon as a semiconductor.
- These solar cells are relatively flexible and can be directly installed with building materials.
- They work great even during clouds when there is low sun light. Here, the disadvantage is that thin-film PV Cells comparatively generate less electricity than crystalline silicon cells.

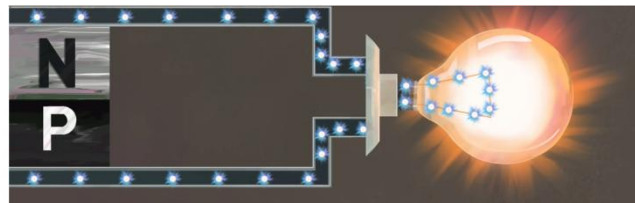
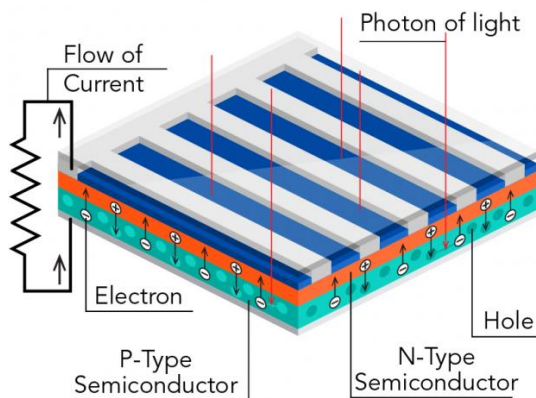
PV Cell

- A PV Cell or Solar Cell or Photovoltaic Cell is the smallest and basic building block of a Photovoltaic System (Solar Module and a Solar Panel).
- These cells vary in size ranging from about 0.5 inches to 4 inches.
- These are made up of solar photovoltaic material that converts solar radiation into direct current (DC) electricity.

- Materials used for photovoltaic include monocrystalline silicon, polycrystalline silicon, microcrystalline silicon, cadmium telluride, and copper indium selenide /sulfide



Photovoltaic Effect



CONSTRUCTION OF SOLAR CELL

A solar cell is basically a junction diode, although its construction is a little bit different from conventional p-n junction diodes. A very thin layer of p-type semiconductor is grown on a relatively thicker n-type semiconductor. Apply a few finer electrodes on the top of the p-type semiconductor layer. These electrodes do not obstruct light to reach the thin p-type layer. Just below the p-type layer there is a p-n junction. It also provides a current collecting electrode at the bottom of the n-type layer. Encapsulate the entire assembly by thin glass to protect the solar cell from any mechanical shock.



WORKING

When light reaches the p-n junction, the light photons can easily enter in the junction, through very thin p-type layer. The light energy, in the form of photons, supplies sufficient energy to the junction to create a number of electron-hole pairs. The incident light breaks the thermal equilibrium condition of the junction. The free electrons in the depletion region can quickly come to the n-type side of the junction. Similarly, the holes in the depletion can quickly come to the p-type side of the junction. Once, the newly created free electrons come to the n-type side, cannot further cross the junction because of barrier potential of the junction.

Similarly, the newly created holes once come to the p-type side cannot further cross the junction because of same barrier potential of the junction. As the concentration of electrons becomes higher in one side, i.e. n-type side of the junction and concentration of holes becomes more in another side, i.e. the p-type side of the junction, the p-n junction will behave like a small battery cell. A voltage is set up which is known as photo voltage. If we connect a small load across the junction, there will be a tiny current flowing through it.

Materials Used in Solar Cell

The materials which are used for this purpose must have band gap close to 1.5eV. Commonly used materials are-

- Silicon.
- GaAs.
- CdTe.
- CuInSe₂

Advantages of photovoltaic systems

- High reliability
- Strong persistence (15-20 year)
- Low maintenance costs
- Zero fuel consumption
- Noise pollution is small
- There is photovoltaic supervision



- Strong security
- Strong independence

Disadvantages of photovoltaic systems

- High startup cost
- Available solar radiation instability
- Have energy storage requirements
- Efficiency needs to be improved
- Lack of knowledge and skills

Photovoltaic Applications

- Solar Farms
- Remote Locations
- Stand-Alone Power
- Power in Space
- Building-Related Needs
- Military Uses
- Transportation

*****The End*****